

1 Learning when to share sunlight: a differentiable
2 simulator and learned tracker control for agrivoltaics
3 on the world’s staple croplands

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6 **Abstract**

Agrivoltaic systems grow crops under photovoltaic (PV) modules and could relieve the competition between food and solar power for land. How much electricity they can produce without compromising food production depends on how sunlight is shared between modules and plants. We built a differentiable agrivoltaic simulator that couples solar geometry, single-axis trackers with backtracking, PV output, crop light and microclimate, a soil-water balance and the SIMPLE crop model with a flowering-stage shade penalty. The PV part agrees with PVGIS at 29 sites ($r = 0.97$, mean bias -4.9%). We trained a neural controller (SALS) by back-propagation through whole seasons to set the daily light-sharing level of the trackers, maximising electricity under a 90% yield-retention floor. In a deployable evaluation, the array density of every site was chosen from its historical seasons with the same rule for all controllers. Performance was measured on geographically held-out sites and years, for 400 crop–site pairs representing the harvested area of wheat, rice, maize, soybean and potato, in baseline and $+1.5$, $+2$ and $+3^\circ\text{C}$ climates. At equal mean yield retention (about 0.905), light-sharing control allowed arrays about three times as dense as static arrays. Electricity rose from 217 to 510 $\text{MWh ha}^{-1} \text{ yr}^{-1}$, and the land-equivalent ratio (LER) from 1.07 to 1.27. The learned controller exceeded the best hand-written rule (phenology rule, LER 1.24) by 0.03–0.05 in LER (about 10% more electricity), at 84–86% of the held-out sites, consistently over three seeds, three spatial splits, three canopy-cooling assumptions and all climates. Its advantage did not depend on stress-aware inputs or on training in warmed climates: the learned policy is a smooth function of crop phenology. The floor was met in 70% of individual seasons and production-weighted retention was below

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0.9 for soybean and maize, so per-site enforcement requires region- and crop-specific design. Equipping 5% of the world’s staple cropland at the floor would yield about 14 PWh yr⁻¹, about half of world generation, but it would need about 10 TWp of modules, 5.7 times the installed global PV capacity. The results quantify how far dynamic control extends agrivoltaics and where the remaining benefit of learning lies.

7 *Keywords:* Agrivoltaics, Differentiable simulation, Food–energy nexus,
8 Solar tracking, Crop modelling, Learned control

9 1. Introduction

10 Two of the defining challenges of this century compete for the same re-
11 source: land. The energy transition requires terawatts of new solar pho-
12 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
13 to two hectares per megawatt. Food production must rise while crops are
14 exposed to more frequent heat waves and droughts. Warming alone is pro-
15 jected to reduce global yields of maize, wheat, rice and soybean by several
16 per cent per degree (Zhao et al., 2017), and a new generation of climate and
17 crop models shows these impacts emerging earlier than previously expected
18 (Jägermeyr et al., 2021). Croplands are also where solar power potential is
19 greatest (Adeh et al., 2019). They are flat, sunny, accessible and close to
20 rural electricity demand, so converting farmland into solar farms is tempting
21 and controversial at the same time.

22 Agrivoltaic (AV) systems combine crops and PV modules on the same
23 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
24 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
25 a hectare can deliver more food plus energy than the two uses side by side,
26 with land-equivalent ratios (LER) well above one in field trials and mod-
27 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
28 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
29 climate. It lowers canopy temperature and evaporative demand, conserves
30 soil moisture and improves water-use efficiency, especially in drylands (Mar-
31 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
32 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a candi-
33 date adaptation measure against heat and drought stress, in addition to a
34 mitigation measure through renewable electricity. The same shade, however,
35 also reduces the light available for photosynthesis. Regulators increasingly

36 require AV systems to retain most of the reference crop yield. The German
 37 pre-standard DIN SPEC 91434, for example, requires at least 66% of the
 38 reference yield (DIN, 2021), and other schemes and many farmers set stricter
 39 floors. How much electricity can be produced without sacrificing food secu-
 40 rity depends on the crop, the local climate, the year and the growth stage.

41 Most AV designs fix the geometry of the array, i.e. the ground coverage
 42 ratio (GCR, module area per ground area) and the tilt or tracking rule, from
 43 generic rules of thumb. Single-axis trackers add a degree of freedom: the
 44 modules can be rotated away from the sun to let light through when the crop
 45 needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the
 46 sun to maximise electricity. Choosing the rotation every day is a sequential
 47 decision problem. The value of light depends on the crop’s phenological stage,
 48 soil water and upcoming weather, and the effect of a decision appears in the
 49 harvest months later. Recent work in this journal has used evolutionary
 50 multi-objective optimisation to lay out PV arrays on farmland (Zhang et al.,
 51 2026). Machine learning for crop management has mainly used reinforcement
 52 learning with black-box simulators (Gautron et al., 2022), which is sample-
 53 inefficient and hard to scale to global, multi-decadal assessments.

54 Two questions follow. First, how much of the electricity that an agri-
 55 voltaic system can produce under a food-security floor depends on the array
 56 density, and how much on dynamic control of the trackers? Second, can a
 57 controller learned from a simulator improve on a well-designed hand-written
 58 rule, and under which conditions? Answering them requires a simulator that
 59 couples the physics of light, PV, microclimate and crop growth, a fair pro-
 60 tocol in which all controllers face the same design constraints, and enough
 61 sites and seasons to separate real effects from noise.

62 This paper makes three contributions. The first is a differentiable agri-
 63 voltaic simulator that couples hourly solar geometry, single-axis trackers with
 64 backtracking, row-to-row shading and PV electricity, the direct and diffuse
 65 light reaching the crop, a microclimate model of canopy temperature and ref-
 66 erence evapotranspiration under the modules, and the SIMPLE crop model
 67 (Zhao et al., 2019) extended with a flowering-stage shade penalty. All com-
 68 ponents are differentiable tensor operations, so a neural controller can be
 69 trained by back-propagation through entire growing seasons (Karniadakis
 70 et al., 2021). The PV part of the simulator is benchmarked against PVGIS.
 71 The second is SALS (stress-aware light sharing), a controller that maps the
 72 daily crop and soil state, the array density and a noisy weather forecast
 73 to a daily light-sharing level, trained to maximise electricity under a yield-

retention floor. The third is a deployable evaluation protocol. The array density of every site is chosen from its historical seasons only, with the same rule for every controller, and performance is measured on held-out sites and years, over three training seeds and three spatial splits.

We apply the protocol to 400 crop–site pairs that represent where maize, wheat, rice, soybean and potato are grown worldwide, sampled in proportion to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al., 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main findings are as follows.

1. Light sharing is the main lever. At the same mean yield retention of 90%, arrays with light-sharing control carry about three times the module area of static arrays and produce 1.8–2.4 times as much electricity (Section 3.1).
2. A learned controller improves on the best hand-written rule by 0.03–0.05 in land-equivalent ratio (about 10% more electricity at equal retention). The gain is consistent over seeds, spatial splits, canopy-cooling assumptions and climates, and it occurs at 84–86% of the held-out sites.
3. The improvement does not come from reacting to heat or drought, and it does not require training in warmed climates. The learned policy is a smooth function of crop phenology (Section 3.4).
4. At the yield floor, agrivoltaics on 5% of the world’s staple cropland could supply about 14 PWh yr^{−1}, but only with about 10 TWp of modules, several times the installed PV capacity of the world (Section 3.8).

2. Materials and methods

2.1. Global cropland sites, weather and crop calendars

We studied five staple crops that together cover about 670 Mha of harvested area and 580 Mha of physical cropland: wheat, rice, maize, soybean and potato. Harvested areas (rainfed and irrigated) were taken from the SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1° cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn without replacement with probability proportional to harvested area. An unweighted average over the sample therefore approximates an area-weighted average over global production. The 400 crop–site pairs lie in 381 distinct cells in 50 countries (Fig. 2). Sowing and maturity dates came from the

GGCMI Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat we used the winter-wheat calendar unless the coldest-month mean temperature was below -8°C , in which case the spring-wheat calendar was used. Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks, 2018): global and diffuse shortwave irradiance, maximum, minimum and dew-point temperature, precipitation and wind speed at 2 m. Each site contributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days after sowing, which gives 7600 site-seasons.

Climate change was represented by the global warming levels $+1.5$, $+2$ and $+3^\circ\text{C}$ relative to 2001–2020. Daily maximum, minimum and dew-point temperatures were shifted uniformly, which keeps relative humidity approximately constant and raises the vapour-pressure deficit. The atmospheric CO_2 concentration was set to 390, 450, 500 and 600 ppm for the baseline and the three warming levels. Precipitation and irradiance were left unchanged. This delta approach isolates the thermal signal of warming; its limitations are discussed in Section 4.4.

2.2. A differentiable agrivoltaic simulator

The simulator advances all site-seasons in parallel with a daily time step for the crop and an hourly time step for radiation and PV (Fig. 1). All equations are implemented as PyTorch tensor operations (Paszke et al., 2019), so that yields and electricity are differentiable with respect to the design and control variables.

Hourly irradiance. Daily global and diffuse irradiance were disaggregated into hourly values with the Collares-Pereira and Rabl (1979) and Liu and Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance on the horizontal is the difference between the two. Solar position follows the standard declination and hour-angle equations.

Array geometry and PV electricity. We consider elevated horizontal single-axis trackers with north–south axes. The axis height is 1.25 module widths, e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The rotation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the rows when the modules are horizontal, which is the stow position during field operations. Let ψ be the solar zenith angle projected onto the east–west vertical plane. The energy-optimal rotation with backtracking (Lorenzo et al.,

2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ otherwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on to the sun, which minimises the shadow on the crop. Given a light-sharing level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

Plane-of-array irradiance combines three terms: beam irradiance on the module, reduced by the fraction of the module shaded by the neighbouring row, $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected light (albedo 0.2). The cell temperature follows the NOCT model (45 °C), with an hourly air temperature interpolated sinusoidally between the daily extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature losses and system losses. The installed capacity is $2.1 \text{ g MWp ha}^{-1}$. A conventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the energy reference E_{PV} .

Validation of the PV model. We compared the simulated specific yield of an isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2 (European Commission, Joint Research Centre, 2022), for a horizontal north-south single-axis tracker with 14% losses, at 30 randomly chosen sites (29 returned data). Across sites, the simulated and PVGIS specific yields were correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit and with the different irradiance databases (NASA POWER vs. SARAH2, ERA5 and NSRDB). Electricity estimates are therefore, if anything, conservative.

Light and microclimate at the crop. Averaged over the row pitch, the fraction of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The fraction of sky diffuse reaching the ground is a two-dimensional isotropic sky-view factor. We pre-computed it for 25 values of g and 13 rotation angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly (differentiably). The daily ratio s of the irradiance at the crop to that of the open field defines the crop's shading. The canopy maximum temperature is lowered by $\kappa(1 - s)$.

κ is uncertain. Field studies under PV modules report small reductions of daytime crop or canopy temperature and little change in air temperature (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds

179 calibrated on air temperature do not transfer directly to canopy temperature
180 (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate
181 $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$
182 for all methods (Section 3.2). Nocturnal warming under the modules is not
183 represented. The reference evapotranspiration under the modules, ET_0 , is re-
184 computed with the FAO-56 Penman–Monteith equation (Allen et al., 1998)
185 from the reduced radiation and canopy temperature. Soil water conserva-
186 tion under the modules (Hassanpour Adeg et al., 2018; Elamri et al., 2018)
187 therefore emerges from the model, but rain redistribution by the modules
188 (Marrou et al., 2013a) is ignored.

189 *Crop growth.* Crop growth follows the SIMPLE model (Zhao et al., 2019)
190 with the species parameters of its Table 1a (Table 1). Daily biomass growth
191 is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

192 where $R_{\text{crop}} = sR$ is the radiation received by the crop. The terms are as
193 follows.

- 194 • f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends
195 only on thermal time, with senescence accelerated by heat and drought
196 stress. Shade therefore reduces the daily biomass increment but not
197 leaf-area development.
- 198 • f_{Heat} decreases linearly between the heat threshold T_{heat} and the ex-
199 treme temperature T_{ext} of the canopy maximum temperature.
- 200 • $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$, where the agricultural reference index for
201 drought (Woli et al., 2012) is computed from a daily soil-water balance
202 with curve-number runoff, drainage and root water uptake. Irrigated
203 fields have $f_{\text{Water}} = 1$.

204 SIMPLE has a fixed harvest index and therefore ignores the high sen-
205 sitivity of grain number to shading around flowering (Fischer, 1985; An-
206 drade et al., 1999). It also ignores the differences in shade tolerance be-
207 tween crop types. In the meta-analysis of Laub et al. (2022), maize and
208 grain legumes lost yield strongly even at low shade levels, whereas C3 ce-
209 reals and tuber crops initially lost less than proportionally; shading trials
210 also show marked effects of shade on maize development (Ramos-Fuentes
211 et al., 2023). We added a stage-dependent penalty. During a critical window

212 $(0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65)$, the harvest index is multiplied by $1 - h\bar{d}$, where $\bar{d}$
 213 is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h
 214 is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al.
 215 (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice
 216 0.5, for which no specific curve was available) and lowest for potato (0.2).
 217 These values are assumptions that reproduce this ranking; they are not cal-
 218 ibrated estimates. Section 3.6 shows the resulting shade-response curves.
 219 Because RUE is linear, SIMPLE does not credit the higher light-use effi-
 220 ciency of shaded canopies under more diffuse light. The penalty, conversely,
 221 makes shading around flowering more damaging than proportional. The net
 222 effect is closer to the field evidence than either assumption alone.

223 The thermal-time requirement to maturity was calibrated for every site as
 224 the mean thermal time between the GGCMI sowing and maturity dates under
 225 the 2001–2020 climate, and the canopy parameters were scaled accordingly.
 226 The cultivar is thus adapted to the present climate, and warming shortens the
 227 season. Growth stops smoothly once the thermal-time requirement is met.
 228 Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site
 229 is represented by a rainfed and an irrigated sub-field in proportion to its
 230 SPAM irrigated fraction. Each sub-field carries its own array, so both yield
 231 *and* electricity are area-weighted over the two sub-fields.

232 2.3. SALS: stress-aware light sharing

233 SALS learns a *control network* $u_t = \pi_\omega(\mathbf{x}_t; g)$ (Algorithm 1). It is a mul-
 234 tilayer perceptron with two hidden layers of 64 units and a sigmoid output,
 235 and it chooses the daily light-sharing level of Eq. (1). Its inputs are the
 236 information available on day t :

- 237 • the crop and soil state: relative thermal time TT/T_{sum} , radiation in-
 238 terception f_{Solar} , relative soil water and the previous day’s ARID;
- 239 • a *forecast* of the day’s maximum temperature (relative to T_{heat} and
 240 T_{opt}), irradiance and ET_0 ;
- 241 • static information: the irrigation status of the sub-field, the array’s
 242 GCR g , latitude, day of season and the crop.

243 Forecasts are perturbed with random errors, independently every day: Gaus-
 244 sian with a standard deviation of 2°C for maximum temperature, and mul-
 245 tiplicative with standard deviations of 20% for irradiance and 15% for ET_0 .

246 The physics always uses the true weather. The same forecast errors are ap-
 247 plied to the rule-based controllers. The controller is conditioned on g . Dur-
 248 ing training, every season is simulated with a GCR drawn uniformly from
 249 $[0.10, 0.50]$, so that one network learns how to control arrays of any density.
 250 Outside the cropping season the trackers backtrack for maximum electricity.

251 For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the
 252 yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional
 253 PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

254 which combines an exact (hinge) and a quadratic penalty for the food-security
 255 floor ρ . Because r_i and e_i are outputs of the differentiable simulator, the
 256 exact gradient of Eq. (3) is obtained by back-propagation through the 365
 257 daily steps of every season (back-propagation through time). This is the key
 258 difference from reinforcement learning with black-box simulators (Gautron
 259 et al., 2022), which estimates gradients from sampled returns.

260 Each minibatch contains 512 site-seasons, each simulated as a rainfed and
 261 an irrigated sub-field, drawn from one of the four climates in turn. Seasons
 262 with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training
 263 and evaluation. The network is trained with Adam (Kingma and Ba, 2015)
 264 (learning rate 3×10^{-3} , cosine decay, gradient-norm clipping at 1) for 300
 265 steps. Growth-stage logistic functions are evaluated in a numerically stable
 266 form, so that gradients remain finite even under extreme heat.

267 2.4. Deployable design selection and baselines

268 A real array has one fixed density, which must be chosen before it is built.
 269 Every method, SALS included, therefore uses the same *history-based design*
 270 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
 271 for which the method’s mean yield retention over the site’s 2001–2014 sea-
 272 sons is at least $\rho = 0.9$. If no value qualifies, no array is built and the site
 273 remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in
 274 at least 80% of the historical seasons. The historical seasons of the test sites
 275 are never used for training, and the weather of the evaluation years is never
 276 used for design. We compare four controllers:

- 277 • *AV static*: backtracking trackers without light sharing ($u = 0$);

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise the control network π_ω
 - 2: **for** step = $1, \dots, N$ **do**
 - 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
 - 4: draw $g_i \sim \mathcal{U}[0.10, 0.50]$; initialise crop and soil states
 - 5: **for** day $t = 1, \dots, 365$ **do**
 - 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
 - 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
 - 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
 - 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}$, $e_i \leftarrow E_i/E_{\text{PV},i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
 - 10: back-propagate through the season; Adam update of ω
-

- 278 • *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the
279 cropping season;
- 280 • *Stress rule*: an expert rule that shares light ($u = 1$) during the main
281 growth phase ($0.1 < TT/T_{\text{sum}} < 0.9$) except on days with a heat-stress
282 forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules
283 track the sun and shade the crop;
- 284 • *SALS*: the learned controller.

285 The open field and a conventional PV plant without crops ($\text{LER} = 1$ each)
286 are the references. Because the design rule is identical for all controllers, dif-
287 ferences between methods reflect the quality of control and the array density
288 that each controller can sustain under the food-security floor. We also com-
289 pare the controllers at identical GCR over the whole grid (matched-density
290 frontiers).

291 2.5. Evaluation protocol, statistics and upscaling

292 Sites were split spatially: $10^\circ \times 10^\circ$ blocks were randomly assigned to train-
293 ing (70%) or test (30%), so that no test site has a training site in the same
294 block. Training uses the 2001–2014 seasons of the training sites. Design
295 selection uses the 2001–2014 seasons of the test sites. Evaluation uses the

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in °C, thermal times in °C d, RUE in gMJ⁻¹; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

2015–2019 seasons of the test sites. The same site-seasons are analysed in all four climates.

We report the following metrics:

- the mean electricity (MWh ha⁻¹ yr⁻¹) and the relative electricity e ;
- the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;
- season-level compliance (share of seasons with $r \geq \rho$) and site-level compliance (share of sites whose mean test-season retention is $\geq \rho$);
- the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011).

Statistical comparisons use site means as the unit of analysis. We report paired Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confidence intervals of the mean difference. SALS was trained with three random seeds on the main split, and with one seed on each of two further spatial splits.

For the global estimate, every site is evaluated in the split in which it is held out. The mean electricity and yield per hectare of each crop are scaled to the *physical* area of that crop in SPAM 2010, which avoids double counting multiple-cropped land (Waha et al., 2020). We report the installed capacity (TWp), the capacity factor and a supply curve.

3. Results

3.1. Electricity and food on held-out sites and years

Table 2 compares the four controllers on the 92 held-out sites of the main split (seasons sown in 2015–2019), after each site’s density has been

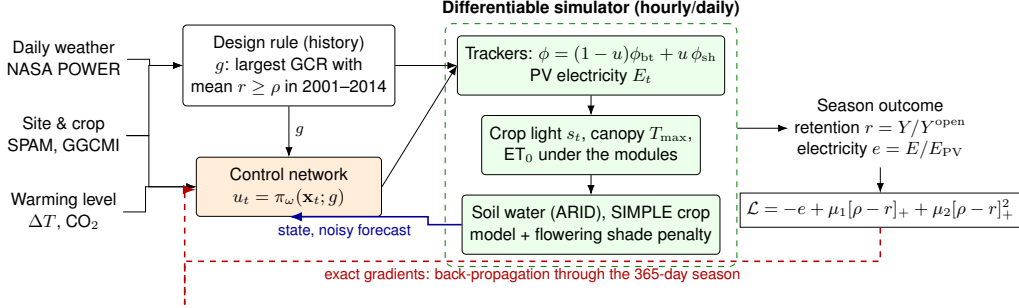


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

319 chosen from its 2001–2014 seasons with the same rule for all controllers. By
 320 construction all methods keep the mean yield retention close to 90% (0.904–
 321 0.917), so differences in electricity are differences in the density that each
 322 controller can sustain under this floor.

323 **Static arrays are the weakest.** Without light sharing, the floor limits
 324 the ground coverage to 6% on average, and arrays produce $217 \text{ MWh ha}^{-1} \text{ yr}^{-1}$,
 325 16% of a conventional PV plant per hectare (LER 1.07). Rotating the
 326 modules to let light through during the cropping season (seasonal shar-
 327 ing) allows a coverage of 17% and $397 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.19). A rule
 328 that shares light only while the crop is growing (phenology rule) reaches
 329 $465 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.24). An expert rule that additionally shades the
 330 crop on forecast heat or drought days (stress rule) is worse than the phenol-
 331 ogy rule (LER 1.18): its density is limited to 12% because it reduces light at
 332 times when the floor is most binding.

333 **The learned controller is best by a modest, consistent margin.**
 334 SALS sustains a density of 17% (similar to the phenology rule), produces
 335 $510 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ and reaches an LER of 1.270 at the same retention. Com-
 336 pared with the phenology rule this is +0.038 in LER (95% CI over sites
 337 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Ta-
 338 ble A.3). Compared with static arrays it is +0.205 (0.176–0.236), with 2.35
 339 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows
 340 the same picture at matched densities: the learned policy lies on or slightly

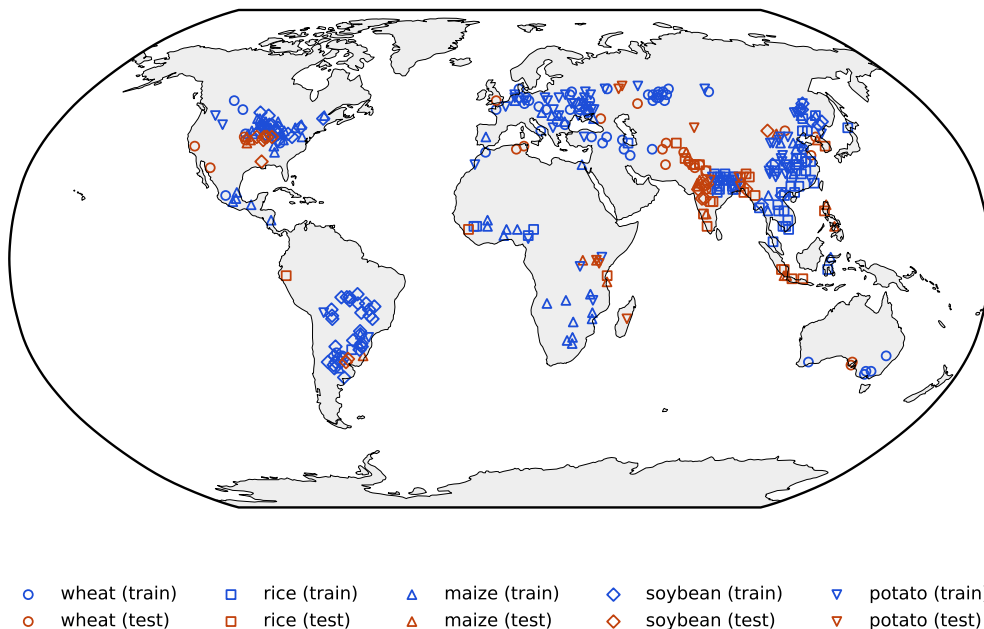


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

341 above the phenology rule over the whole density range, both lie well above
342 seasonal sharing and the stress rule, and static arrays are far below.

343 **What the floor does and does not guarantee.** Enforcing the floor
344 on the historical mean does not guarantee it in every season. In the test
345 seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule
346 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%).
347 The lower tail of SALS is therefore slightly heavier than that of the rules:
348 its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring
349 $r \geq 0.9$ in at least 80% of a site’s historical seasons (chance-constrained
350 design) raises the season-level compliance to 90–94% for the light-sharing
351 controllers. It costs about 0.065 in LER for SALS (1.205), which remains the
352 best controller (phenology rule 1.192; Section 3.2).

353 3.2. Robustness

354 **Seeds and splits.** Three training seeds of SALS give LERs of 1.270,
355 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at

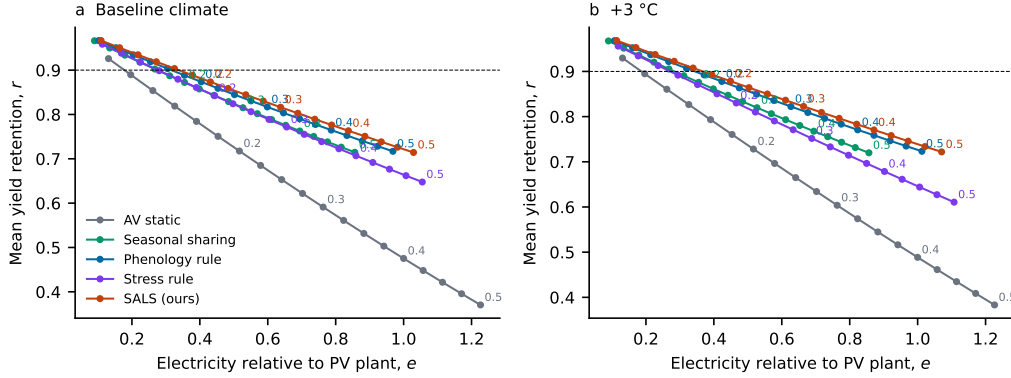


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

+3 °C. On two further spatial splits, in which different blocks of sites are held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under the baseline climate and by 0.052 and 0.039 at +3 °C (main split: 0.034 and 0.032). In every split SALS ranks first, the phenology rule second and static arrays last.

Canopy cooling. The canopy-cooling coefficient κ is the most uncertain microclimate parameter. We repeated the whole protocol, including the historical design step, with $\kappa = 0, 1$ and 2 °C. Electricity changes with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and 549 MWh ha⁻¹ yr⁻¹ under the baseline climate), but the retention is held at 0.905 by the design rule and the ranking is unchanged. SALS exceeds the phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate (0.022, 0.032 and 0.051 at +3 °C).

Which ingredients matter. Two ablations test the design of SALS at the same training budget. Masking the stress inputs (soil water, ARID, and the maximum temperature, irradiance and ET₀ forecasts) leaves the LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297 at +3 °C). Training only on the baseline climate gives 1.268 and 1.291. The improvement over the rules therefore does not depend on stress-aware inputs or on training in warmed climates.

3.3. Additional experiments

TO BE FILLED

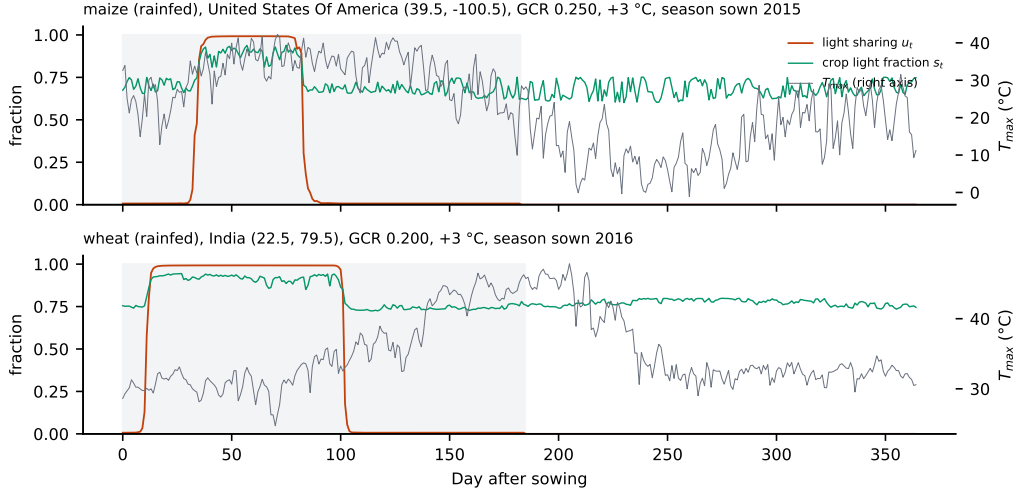


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

3.4. What the learned policy does

Averaged over the rainfed test seasons of the main split at a density of 0.3, the light-sharing level of SALS is a smooth function of relative thermal time. It is close to zero in the first 10% of the thermal time required for maturity, rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and 70%, and falls through 0.58 at 80–90% to about zero at maturity. The plateau covers the window in which our crop model penalises shading through the harvest index. The ramps at both ends let the modules produce electricity when additional light is worth least to the crop. Fig. 4 shows two example seasons. The phenology rule shares light during the whole growing period. SALS gains its advantage by yielding light when it matters and collecting electricity at the edges of the season.

The policy also shares less light on days with a forecast heat stress: the mean level is 0.35 on such days against 0.60 (baseline) and 0.51 ($+3^{\circ}\text{C}$) on other days, and 58–63% of sites show a lower level on heat days. This cannot be read as a stress response, however. Heat days occur late in the season, when the phenological ramp is already falling, and a policy without any stress input shows a similar contrast (0.37 vs 0.57 in a separate check). The ablation above confirms that stress information does not improve the

397 result.

398 3.5. Crops and regions

399 The benefit of light sharing differs between crops (baseline climate, SALS).
400 Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and
401 577 MWh ha⁻¹ yr⁻¹; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31),
402 and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20
403 for both). The stress rule has a lower LER for every crop (1.26 for soybean,
404 1.14–1.21 for the others).

405 The floor is not met evenly. Production-weighted retention is 0.888 for
406 soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose
407 more than low-yielding ones. Regionally, Latin America and Oceania (4 sites)
408 combine the highest density (GCR 0.41) and electricity (1090 MWh ha⁻¹ yr⁻¹)
409 with a production-weighted retention of only 0.862. Africa and the Middle
410 East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention
411 of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902
412 (67% of sites) at a density of 0.14. A regulatory floor enforced per site would
413 therefore need a region- and crop-specific design step, not a global one.

414 3.6. Model behaviour and validation

415 **PV output.** The simulated specific yield of an isolated tracker row agrees
416 with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29
417 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute deviation
418 of 5.5% (Section 2.2).

419 **Shade response.** Fig. 5 shows the simulated relative yield of rainfed
420 crops under static arrays as a function of the season-mean shading of the crop.
421 Yield declines close to proportionally with shading: 84–89% relative yield at
422 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat
423 loses most and potato and maize least. This response is steeper than the *less*
424 *than proportional* losses that Laub et al. (2022) report for small shade levels
425 of C3 cereals and tubers, and it does not reproduce their finding that maize
426 and grain legumes are more shade sensitive than C3 cereals. In the simulator,
427 shading also relieves heat and drought in rainfed maize and soybean, which
428 offsets the higher harvest-index penalty we assigned to them. The absolute
429 effect of shading on each crop is therefore uncertain (Section 4.4).

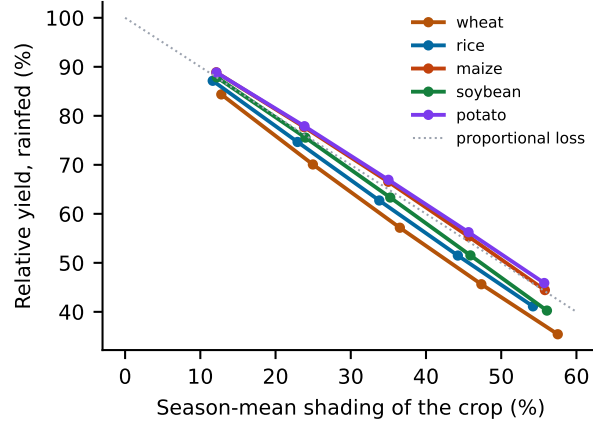


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

3.7. Agrivoltaics under warming

Under warming, the open-field production of the held-out sites (Fig. 6) falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soybean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production with light-sharing control follows the open field at about 90% of its level, so it inherits the same climate impacts (maize −31%, rice −27% relative to the baseline open field at +3 °C). Agrivoltaics does not protect staple production against warming in this model. The LER of light-sharing systems increases slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265), because the floor allows slightly denser arrays in warmer conditions (GCR 0.173 to 0.177 and electricity 510 to 540 MWh ha^{−1} yr^{−1}). The margin of SALS over the rules does not grow with warming (Table A.3).

3.8. Global potential and what it would take

Every site is evaluated when it is held out (219 of the 400 crop-site pairs, in at least one split), and each crop’s mean electricity per hectare is scaled to its physical area in SPAM 2010 (581 Mha for the five crops, which avoids counting multiple-cropped land twice). Equipping 5% of this area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr^{−1} under the baseline climate and 16.0 PWh yr^{−1} at +3 °C, with a production-weighted retention of 0.903 and 0.900. The phenology rule would give 13.0

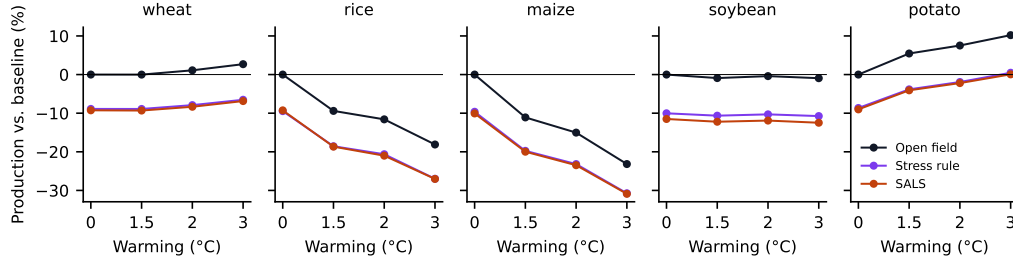


Figure 6: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

and 14.5 PWh yr^{-1} , and static arrays 6.4 and 6.9 PWh yr^{-1} . For comparison, world electricity generation was about 30 PWh in 2023 (Ember, 2024).

These numbers require large amounts of hardware. SALS at 5% coverage needs 10.6 TWp of modules (11.3 TWp at $+3^\circ\text{C}$), at a capacity factor of 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW (IRENA, 2025), so the requirement is about 5.7 times the installed PV fleet. The electricity is also variable, peaks at midday and would be curtailed or stored. The figures are technical potentials, not projections. Ordering sites by electricity per hectare gives a supply curve (Fig. 7): the best 2% of the staple cropland would supply 16 PWh yr^{-1} and the best 5% 40 PWh yr^{-1} with SALS. The best tenth of the sites are mostly rainfed maize and soybean sites in the Americas and in southern and eastern Africa (7% mean irrigated fraction against 29% overall), which are also among the regions where the floor is hardest to meet (Section 3.5). Siting by electricity alone would therefore concentrate agrivoltaics where food-security risk is highest.

4. Discussion

4.1. What the results establish

Three findings stand on consistent evidence. First, sharing light with the crop is the main lever on the electricity that agrivoltaic arrays can produce under a yield floor. With static trackers the floor limits arrays to about 6% ground coverage and an LER of 1.07. Any light-sharing control allows about three times the density, and the LER rises to 1.18–1.24 for hand-written rules. Second, a controller learned through the differentiable simulator is better than the best rule we could write, by 0.03–0.05 in LER and about 10% in electricity at equal retention. The gain is small compared with the

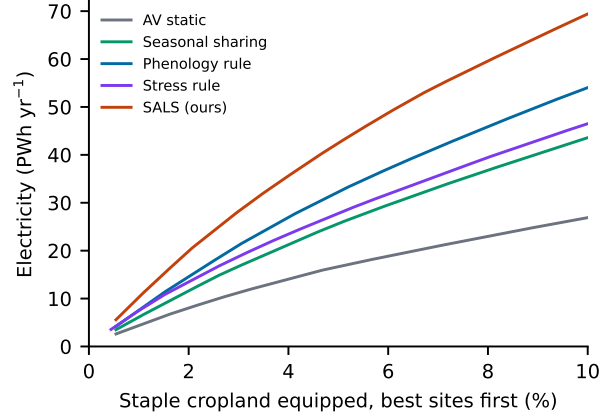


Figure 7: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

gap between static and dynamic control. It is, however, reproducible over training seeds (range 0.002 in LER), three spatial splits, three canopy-cooling assumptions and four climates, and it is found at 84–86% of the held-out sites. Third, the learned controller is a smooth function of crop phenology. It shares light almost fully across the middle of the season, and it produces electricity at the beginning and the end. Our ablations show that it does not use stress information, and that it does not need training in warmed climates, to achieve the gain.

The modest size of the learned gain should be stated as a result. The phenology rule already contains the main idea, and the learned schedule refines its timing. A reader who needs a simple, auditable controller can implement the phenology rule and obtain about 85% of the electricity gain of SALS over static arrays. What the learned controller adds is the option to optimise the schedule automatically for other objectives, crops or physics without designing new rules. The experiments on conditioning by the food-security floor and on forecast errors probe this flexibility (Sections 3.2 and 3.3).

4.2. What the results do not establish

The early hypothesis that stress-aware control, that is, shading the crop on heat and drought days, would be a main source of benefit under warming

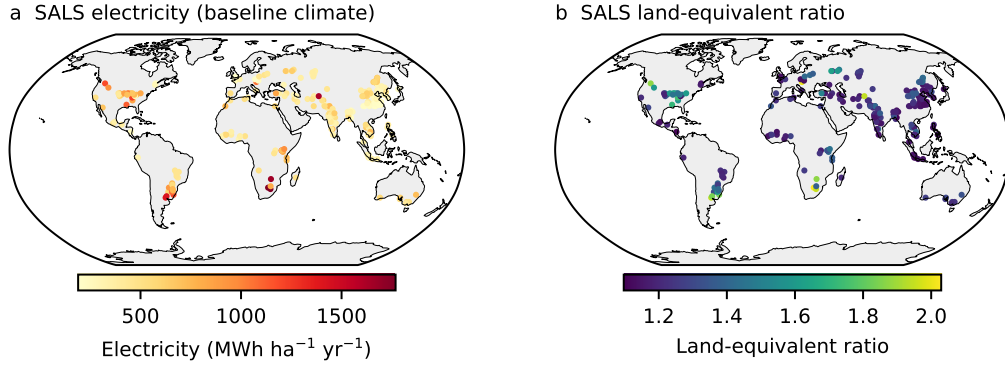


Figure 8: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

was not supported. The stress rule performs worse than the phenology rule in every climate, and a learned controller without stress inputs performs as well as one with them. A plausible reason is that in our crop model the yield gain from shading on a hot or dry day is smaller than the loss of light on that day, so stress information does not help. Whether this holds for crop models with more detailed canopy-temperature and heat-stress responses is an open question. Agrivoltaics does not protect staple production against warming in our simulations: light-sharing systems produce about 90% of the open-field yield in every climate and follow the same decline in maize and rice (Fig. 6).

4.3. Food security in practice

Holding the historical mean retention at 90% does not hold it in each season or at each site. 70% of SALS seasons meet the floor, the lower tail is slightly heavier than for the rules, and production-weighted retention is below 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-constrained design raises season-level compliance to about 92% at a cost of 0.065 in LER, and the ranking of the controllers is preserved. Whichever formulation is chosen, a per-site regulatory floor would need a region- and crop-specific design step. The upscaled potential (Section 3.8) shows the trade-off in magnitude. The 14 PWh yr^{-1} that 5% of the staple cropland could supply need about 10 TWp of modules, several times the present global fleet, and the best sites are rainfed maize and soybean areas where production-weighted retention is lowest.

518 4.4. Limitations

- 519 • *Simulation only.* All results come from a simulator. PV output was
520 benchmarked against PVGIS (29 sites, mean bias -4.9%), but crop
521 yield under shade was not validated against trial data. The simulated
522 shade response is close to proportional and does not reproduce the
523 ranking of shade tolerance among crop types reported in meta-analyses
524 (Laub et al., 2022). The harvest-index sensitivities h are assumptions.
- 525 • *Canopy cooling.* The effect of shade on canopy temperature was repre-
526 sented by one coefficient κ , applied to a heat-stress function that was
527 calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018).
528 The ranking of the controllers is robust to κ between 0 and 2°C , but
529 the absolute electricity changes by about $\pm 8\%$. Nocturnal warming
530 under modules and rain redistribution are not modelled.
- 531 • *Crop model.* SIMPLE has a fixed harvest index and linear radiation-
532 use efficiency. More detailed models with grain-number formation and
533 canopy energy balance may favour different schedules.
- 534 • *Climate scenarios.* Warming was applied as uniform temperature shifts
535 with fixed precipitation, radiation and relative humidity, and with un-
536 changed cultivars and sowing dates. It is not a bias-adjusted climate
537 projection.
- 538 • *Forecasts.* The controllers saw the actual state of the crop and soil plus
539 a noisy forecast of the day’s weather, not a real forecast system with
540 sensor and model errors.
- 541 • *Economics and deployment.* There is no cost, market or grid analysis.
542 The value of midday electricity, curtailment, storage, structural cost
543 and farmer acceptance (Trommsdorff et al., 2021) will determine the
544 realisable share. The average row pitch at the chosen densities (about
545 13 m for 2.2-m modules) leaves room for machinery, but the densest
546 designs (GCR up to 0.5) have a pitch of only 4.4 m.

547 5. Conclusions

548 A differentiable simulator of trackers, PV output, crop light, microclimate
549 and crop growth allowed us to compare agrivoltaic controllers fairly on 400

crop–site pairs, with the array density of each site chosen from its own history. At a mean yield retention of 90%, light-sharing control allows about three times the module density of static arrays, raising the land-equivalent ratio from 1.07 to 1.18–1.24 with hand-written rules and to 1.27 with a learned controller. The learned controller improves on the best rule by 0.03–0.05 in LER (about 10% more electricity), reproducibly over seeds, spatial splits, canopy-cooling assumptions and climates. The improvement comes from a smooth phenology-dependent schedule, not from reacting to heat or drought. The yield floor is met on average but not in every season or region, and the electricity potential of 5% of staple cropland (about 14 PWh yr^{−1}) requires about 10 TWp of modules. Future work should validate the crop response to shade under trackers with field data, extend the crop model to grain-number formation and canopy energy balance, add costs and grid constraints, and pursue controllers that guarantee the food-security floor per site and per season.

CRedit authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al., 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA POWER daily weather (Sparks, 2018) and the SIMPLE crop parameters (Zhao et al., 2019). The code of the simulator, of SALS and of all experiments, together with the site list, is available from the author’s public repository (<https://github.com/vinhqdang/Artificial-Intelligence-in-Agriculture>).

580 **Funding**

581 This research did not receive any specific grant from funding agencies in
582 the public, commercial, or not-for-profit sectors.

583 **Appendix A. Statistical comparison**

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Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS (ours)	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS (ours)	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS (ours)	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS (ours)	0.18	540	0.39	0.907	0.904	68	75	1.30

Table A.3: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). ΔLER : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	ΔLER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92